EXP -3.JENKINS SETUP ON AWS

Launch the two instances on aws

Give the names to the instances

☐	devm	i-01fe94a504395bfe3	Running	t2.micro	Initializing	View alarms +
☐	sitm	i-06bbcd7c3e27800b4	Running	t2.micro	Initializing	View alarms +

JENKINS SETUP:

Connect to dev machine through git bash

Update the apt repository
Sudo apt-get update

```
ubuntu@ip-172-31-15-229: ~
ubuntu@ip-172-31-15-229:~$ sudo apt-get update
```

Download java

Sudo apt-get install -y openjdk-21-jdk

```
ubuntu@ip-172-31-15-229:~$ sudo apt-get install -y openjdk-21-jdk
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  adwaita-icon-theme alsa-topology-conf alsa-ucm-conf at-spi2-core ca-certificates-java dconf-gsettings-backend dconf-service fontconfig fontconfig-config
  fonts-dejavu-core fonts-dejavu-extra gsettings-desktop-schemas gtk-update-icon-cache hicolor-icon-theme humanity-icon-theme java-common libasound2 libasound2-data
  libatk-bridge2.0-0 libatk-wrapper-java libatk-wrapper-java-jni libatk1.0-0 libatk1.0-data libatspi2.0-0 libavahi-client3 libavahi-common-data libavahi-common3
  libcairo-gobject2 libcairo2 libcolord2 libcups2 libdatrie1 libdconf1 libdeflate0 libdrm-amdgpu1 libdrm-intel1 libdrm-nouveau2 libdrm-radeon1 libepoxy0 libfontconfig1
  libfontenc1 libgdk-pixbuf-2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgif7 libgl1 libgl1-amber-dri libgl1-mesa-dri libglapi-mesa libglvnd0 libglx-mesa0
  libglx0 libgraphite2-3 libgtk-3-0 libgtk-3-bin libgtk-3-common libharfbuzz0b libice-dev libice6 libjbig0 libjpeg-turbo8 libjpeg8 liblcms2-2 libllvm15 libpango-1.0-0
  libpangocairo-1.0-0 libpangoft2-1.0-0 libpciaccess0 libpcsclite1 libpixman-1-0 libpthread-stubs0-dev libraqm0 librsvg2-2 librsvg2-common libsensors-config libsensors5 libsm-dev
  libsm6 libthai-data libthai0 libtiff5 libwayland-client0 libwayland-cursor0 libwayland-egl1 libwebp7 libx11-dev libx11-xcb1 libxau-dev libxaw7 libxcb-dri2-0
  libxcb-dri3-0 libxcb-glx0 libxcb-present0 libxcb-randr0 libxcb-render0 libxcb-shape0 libxcb-shm0 libxcb-sync1 libxcb-xfixes0 libxcb1-dev libxcomposite1 libxcursor1
  libxdamage1 libxdmcp-dev libxfixes3 libxft2 libxi6 libxinerama1 libxkbcommon0 libxkbfile1 libxmu6 libxpm4 libxrandr2 libxrender1 libxshmfence1 libxt-dev libxt6 libxtst6
  libxv1 libxxf86dga1 libxxf86vm1 openjdk-21-jdk-headless openjdk-21-jre openjdk-21-jre-headless session-migration ubuntu-mono x11-common x11-utils xkb-data-dev
  xorg-sgml-doctools xtrans-dev
```

Open the bowser type jenkins.io in the url --> click on the download

Right click on the generic java package --> copy the link

https://get.jenkins.io/war-stable/2.492.2/jenkins.war

Run the below command in the gitbash dev terminal

$wget https://get.jenkins.io/war-stable/2.492.2/jenkins.war

Start the jenkins

$java -jar jenkins.war

Administration password is available on the terminal

ACCESSING JENKINS:

Open the browser -->Take the public ip of devserver :8080

Copy paste the admin password

Click on install suggested plugin

Getting Started

Customize Jenkins

Plugins extend Jenkins with additional features to support many different needs.

Install suggested plugins

Install plugins the Jenkins community finds most useful.

Select plugins to install

Select and install plugins most suitable for your needs.

Jenkins 2.492.2

Getting Started

Create First Admin User

Username

admin

Password

•••••

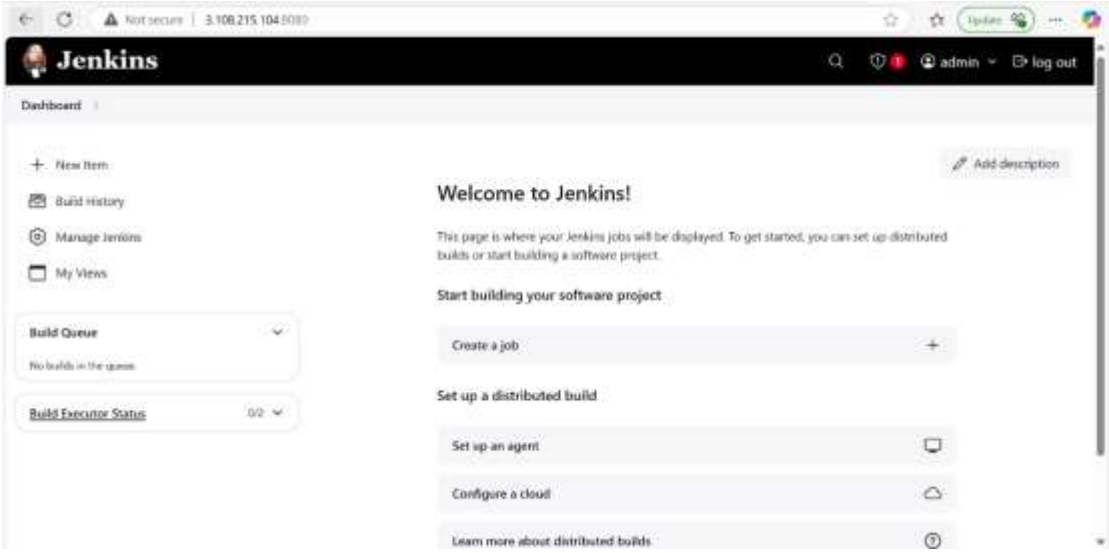
Confirm password

•••••

Jenkins 2.492.2

Skip and continue as admin

Save and Continue



EXP – 4 Build war file:

Connect dev terminal through git

Check java is install or not. If not install then install java

Check maven is install or nto . if not install then install maven

Check jenkins is install or not . if not install then install jenkins

Maven download:

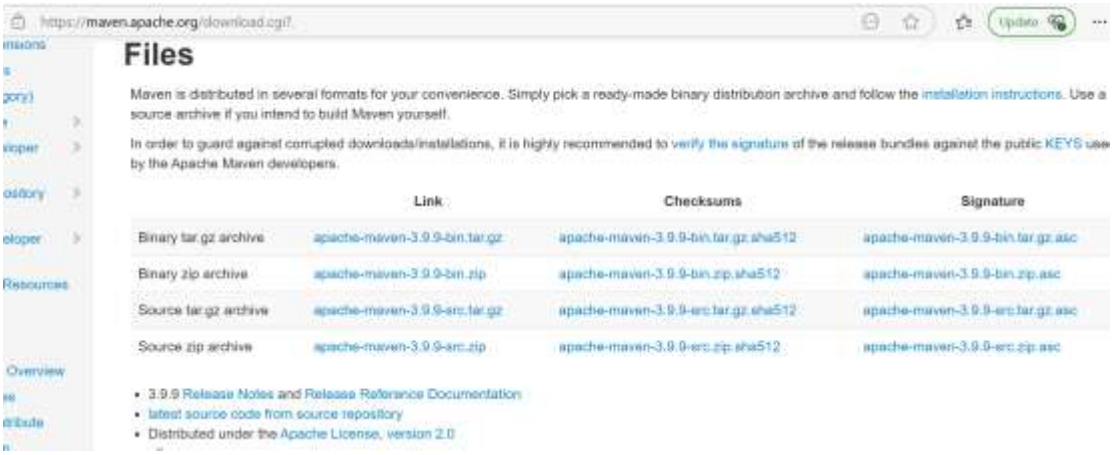
Go to the root directory

Sudo su -

Cd /opt

```
ubuntu@ip-172-31-15-229:~$ sudo su -
root@ip-172-31-15-229:~# cd /opt
root@ip-172-31-15-229:/opt#
```

Open the browser type maven download



Right click on the apache-maven.3.9.9.bin.tar.gz -->copy the link

Wget https://dlcdn.apache.org/maven/maven-3/3.9.9/binaries/apache-maven-3.9.9-bin.tar.gz

Unzip the maven gip file

Tar -xvzf `apache-maven-3.9.9-bin.tar.gz`

```
root@ip-172-31-15-229:/opt# ll
total 8900
drwxr-xr-x  2 root root    4096 Mar 10 13:33 ./
drwxr-xr-x 19 root root    4096 Mar 10 13:06 ../
-rw-r--r--  1 root root 9102945 Aug 17  2024 apache-maven-3.9.9-bin.tar.gz
root@ip-172-31-15-229:/opt# tar -xvzf apache-maven-3.9.9-bin.tar.gz
apache-maven-3.9.9/README.txt
```

#ll

```
root@ip-172-31-15-229:/opt# ll
total 8904
drwxr-xr-x  3 root root    4096 Mar 10 13:35 ./
drwxr-xr-x 19 root root    4096 Mar 10 13:06 ../
drwxr-xr-x  6 root root    4096 Mar 10 13:35 apache-maven-3.9.9/
-rw-r--r--  1 root root 9102945 Aug 17  2024 apache-maven-3.9.9-bin.tar.gz
root@ip-172-31-15-229:/opt#
```

Rename apache-maven-3.9.9 to maven

```
root@ip-172-31-15-229:/opt# mv apache-maven-3.9.9 maven
root@ip-172-31-15-229:/opt# ll
total 8904
drwxr-xr-x  3 root root    4096 Mar 10 13:37 ./
drwxr-xr-x 19 root root    4096 Mar 10 13:06 ../
-rw-r--r--  1 root root 9102945 Aug 17  2024 apache-maven-3.9.9-bin.tar.gz
drwxr-xr-x  6 root root    4096 Mar 10 13:35 maven/
root@ip-172-31-15-229:/opt#
```

Cd maven

#ll

```
root@ip-172-31-15-229:/opt# cd maven
root@ip-172-31-15-229:/opt/maven# ll
total 56
drwxr-xr-x 6 root root  4096 Mar 10 13:35 ./
drwxr-xr-x 3 root root  4096 Mar 10 13:37 ../
-rw-r--r-- 1 root root 18920 Aug 14  2024 LICENSE
-rw-r--r-- 1 root root  5034 Aug 14  2024 NOTICE
-rw-r--r-- 1 root root  1279 Aug 14  2024 README.txt
drwxr-xr-x 2 root root  4096 Mar 10 13:35 bin/
drwxr-xr-x 2 root root  4096 Mar 10 13:35 boot/
drwxr-xr-x 3 root root  4096 Aug 14  2024 conf/
drwxr-xr-x 4 root root  4096 Mar 10 13:35 lib/
root@ip-172-31-15-229:/opt/maven#
```

Note down the maven path
#pwd

```
/opt/maven
```

Move to cd bin

#ll

Note down the maven bin path (/opt/maven/bin)
#pwd

```
root@ip-172-31-15-229:/opt/maven# cd bin
root@ip-172-31-15-229:/opt/maven/bin# ll
total 40
drwxr-xr-x 2 root root 4096 Mar 10 13:35 ./
drwxr-xr-x 6 root root 4096 Mar 10 13:35 ../
-rw-r--r-- 1 root root  327 Aug 14  2024 m2.conf
-rwxr-xr-x 1 root root 5917 Aug 14  2024 mvn*
-rw-r--r-- 1 root root 6324 Aug 14  2024 mvn.cmd
-rwxr-xr-x 1 root root 1684 Aug 14  2024 mvnDebug*
-rw-r--r-- 1 root root 2169 Aug 14  2024 mvnDebug.cmd
-rwxr-xr-x 1 root root 1611 Aug 14  2024 mvnyjp*
root@ip-172-31-15-229:/opt/maven/bin# pwd
/opt/maven/bin
root@ip-172-31-15-229:/opt/maven/bin#
```

Now check the maven is install or not

Mvn --version

```
root@ip-172-31-15-229:/opt/maven/bin# mvn --version
Command 'mvn' not found, but can be installed with:
apt install maven
root@ip-172-31-15-229:/opt/maven/bin#
```

Try with the below command

```
./mvn --version
```

```
root@ip-172-31-15-229:/opt/maven/bin# ./mvn --version
Apache Maven 3.9.9 (8e8579a9e76f7d015ee5ec7bfcdc97d260186937)
Maven home: /opt/maven
Java version: 21.0.6, vendor: Ubuntu, runtime: /usr/lib/jvm/java-21-openjdk-amd64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.8.0-1021-aws", arch: "amd64", family: "unix"
root@ip-172-31-15-229:/opt/maven/bin#
```

Now go back to the roor directory --> cd /

Check the maven version --> mvn --version

It shows maven is not found.

```
root@ip-172-31-15-229:/# mvn --version
Command 'mvn' not found, but can be installed with:
apt install maven
root@ip-172-31-15-229:/#
```

So we need to create environment variable

Go to root directory with below command

Cd ~

```
root@ip-172-31-15-229:/# cd ~
root@ip-172-31-15-229:~# pwd
/root
root@ip-172-31-15-229:~#
```

#ll

U can see the .profile file

Open the file with below command

Sudo vim .profile

```
# ~/.profile: executed by Bourne-compatible login shells.

if [ "$BASH" ]; then
  if [ -f ~/.bashrc ]; then
    . ~/.bashrc
  fi
fi

mesg n 2> /dev/null || true
~
```

Go to the insert mode(click on I) and give the maven,java home and m2 paths here

```
# ~/.profile: executed by Bourne-compatible login shells.

if [ "$BASH" ]; then
  if [ -f ~/.bashrc ]; then
    . ~/.bashrc
  fi
fi
M2_HOME=/opt/maven
M2=/opt/maven/bin
JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64
PATH=$PATH:$HOME/bin:$JAVA_HOME:$M2_HOME:$M2

mesg n 2> /dev/null || true
```

:wq

To get the java_home path
#find / -name java-21* (`/usr/lib/jvm/java-21-openjdk-amd64`)

```
root@ip-172-31-15-229:~# find / -name java-21*
/usr/share/gdb/auto-load/usr/lib/jvm/java-21-openjdk-amd64
/usr/lib/jvm/java-21-openjdk-amd64
/usr/lib/debug/usr/lib/jvm/java-21-openjdk-amd64
/etc/java-21-openjdk
```

#echo $PATH

```
root@ip-172-31-15-229:~# echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
root@ip-172-31-15-229:~#
```

U CANT SEE THE JAVA AND MAVEN PATH ABVOE . SO WE NEED TO RESTART THE .PROFILE FILE with below command

#source .profile

```
root@ip-172-31-15-229:~# source .profile
root@ip-172-31-15-229:~# echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/root/bin:/usr/lib/jvm/java-21-openjdk-amd64:/opt/maven:/opt/maven/bin
root@ip-172-31-15-229:~#
```

Now u can see the java and maven paths

Now check the maven version

#mvn --version -->now u can able to see the maven

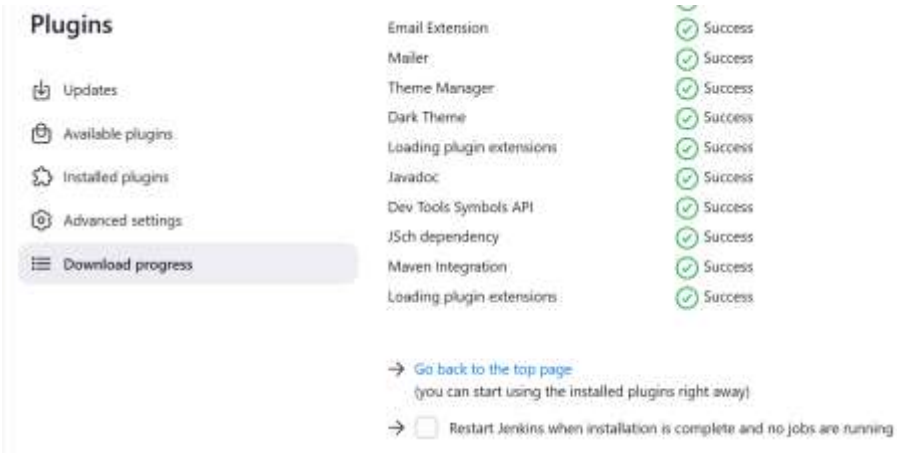
```
root@ip-172-31-15-229:~# mvn --version
Apache Maven 3.9.9 (8e8579a9e76f7d015ee5ec7bfcdc97d260186937)
Maven home: /opt/maven
Java version: 21.0.6, vendor: Ubuntu, runtime: /usr/lib/jvm/java-21-openjdk-amd64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.8.0-1021-aws", arch: "amd64", family: "unix"
root@ip-172-31-15-229:~# |
```

Now go to the jenkins dashboard u need to install one plugin (maven integratin)

Managejenkins-->plugins-->available plugins-->maven integration plugin

Without this plugin we can't able to see the maven project

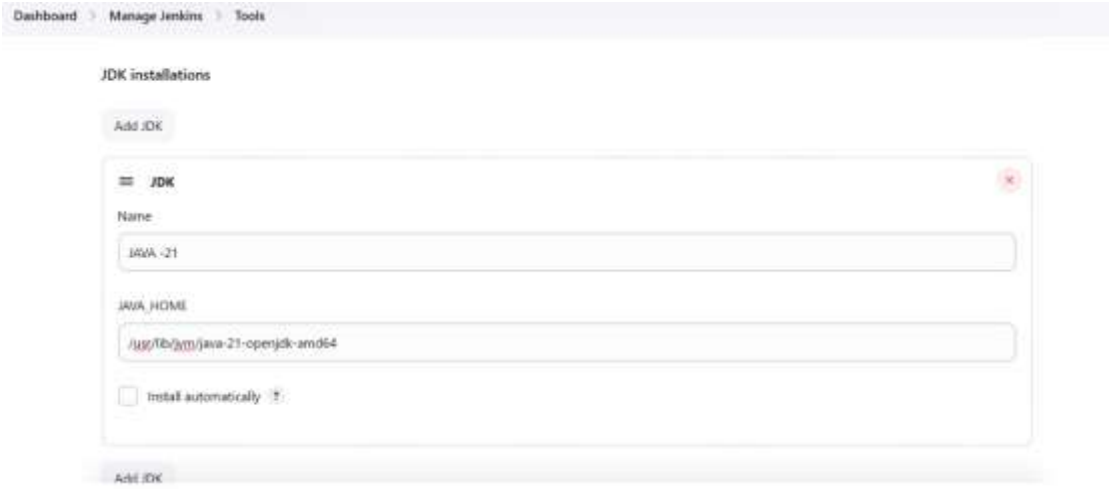
Once installed , click on the restart jenkins



U need to add java and maven paths in the jenkins

Go to jenkins dashboard-->managejenkins-->tools

Add JDK and add MAVEN



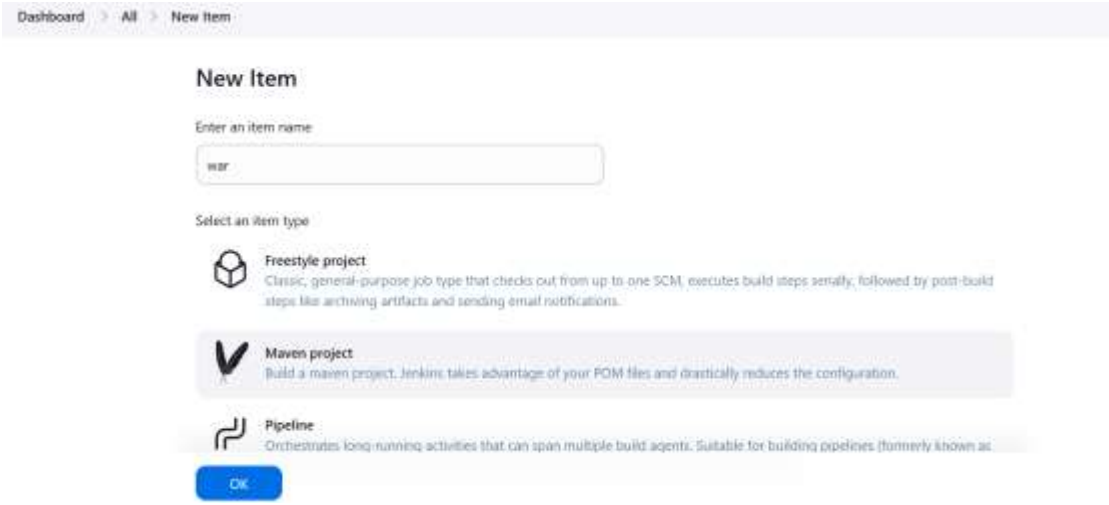
UNCHECK THE install automatically

Add maven

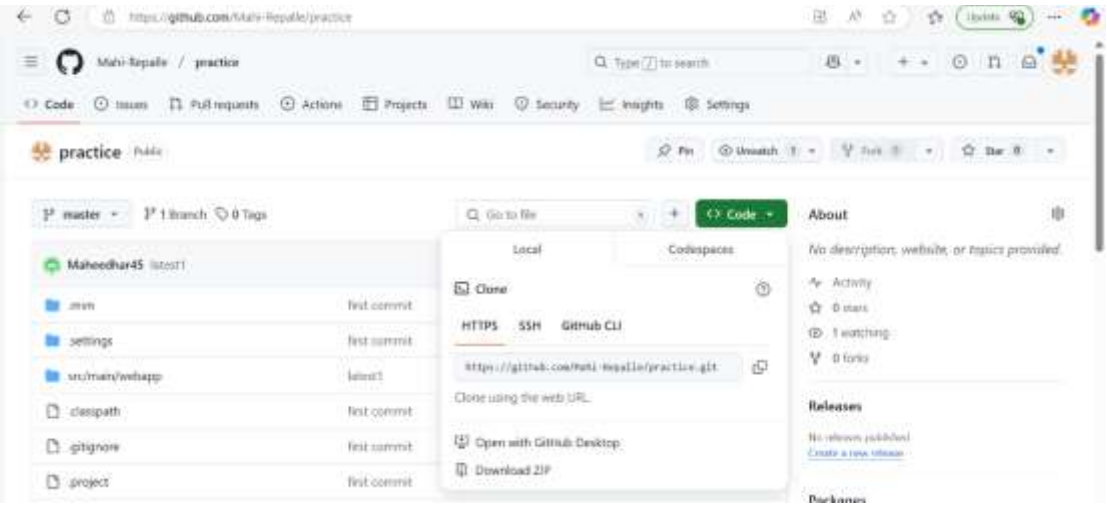
Click on apply and save it

Now we can create job for war file

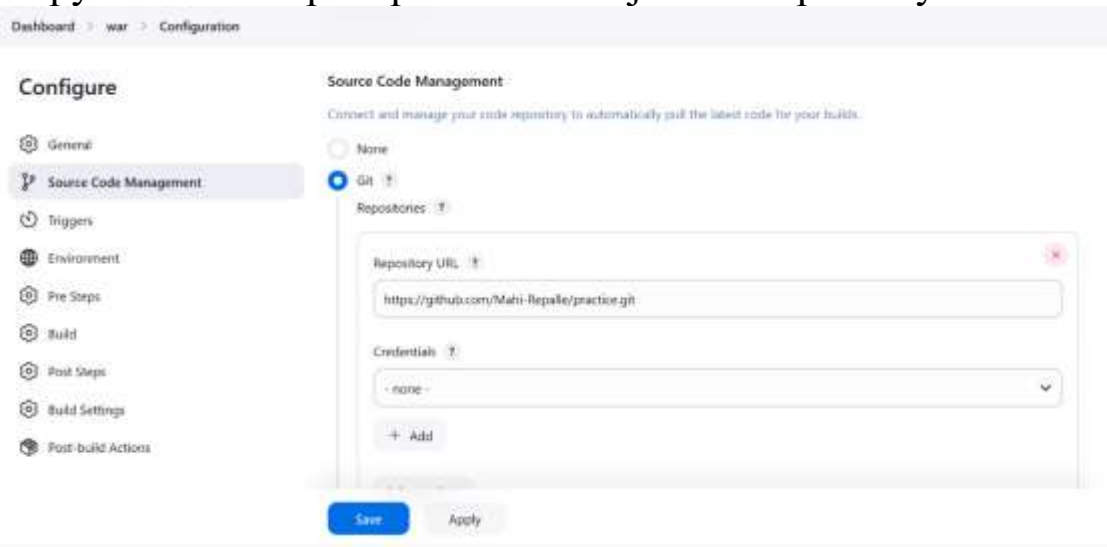
Dashboard-->newitem-->war(give any name) select maven project click on ok



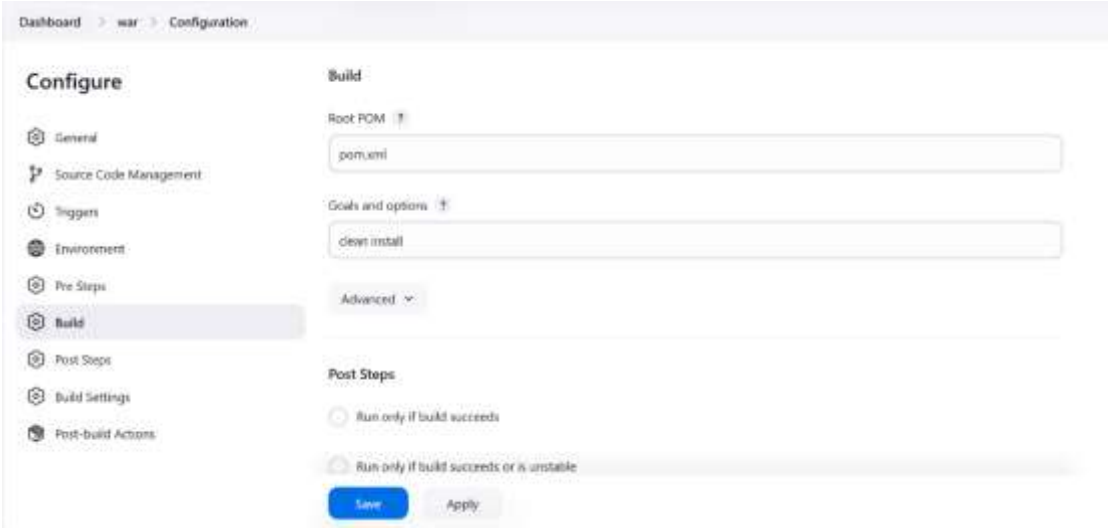
Source code management-->select git--> give the github url project path(open my github account-->go to the repositories
Select the practice repository)



Copy the HTTPS path paste it in the jenkins repository url

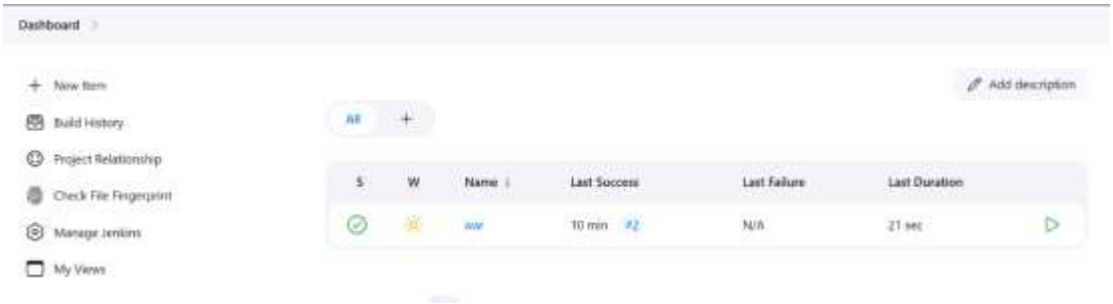


Select the build goal and options write the command clean install click on apply and save



Click on build now

If job execution is taking lot of time
Then
logout the jenkins
Stop the dev instance
Restart the dev instanc
Start the jenkins server
Run the job
Now ur job will be excuted



```
Started by user admin
Running as SYSTEM
Building in workspace /home/ubuntu/.jenkins/workspace/war
The recommended git tool is: NONE
No credentials specified
 > git rev-parse --resolve-git-dir /home/ubuntu/.jenkins/workspace/war/.git # timeout=10
Fetching changes from the remote Git repository
 > git config remote.origin.url https://github.com/Mahi-Repalle/practice.git # timeout=10
Fetching upstream changes from https://github.com/Mahi-Repalle/practice.git
```

```
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  10.805 s
[INFO] Finished at: 2025-03-10T14:36:37Z
[INFO] ------------------------------------------------------------------------
Waiting for Jenkins to finish collecting data
[JENKINS] Archiving /home/ubuntu/.jenkins/workspace/war/pom.xml to DevOpsCourse/MyDevPipeline/0.0.1-
SNAPSHOT/MyDevPipeline-0.0.1-SNAPSHOT.pom
[JENKINS] Archiving /home/ubuntu/.jenkins/workspace/war/target/MyDevPipeline.war to
DevOpsCourse/MyDevPipeline/0.0.1-SNAPSHOT/MyDevPipeline-0.0.1-SNAPSHOT.war
channel stopped
Finished: SUCCESS
```